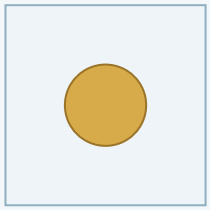
DATASET / GEOMETRY ONLY

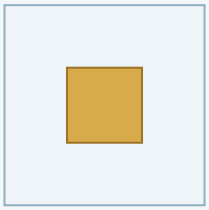
Geometric design space

Four Au shape families. Full 0–320 nm domain, without imposed minimum dimensions.



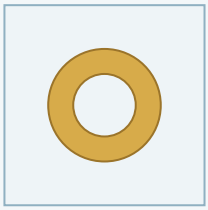
CIRCULAR DISK

D = 130 nm



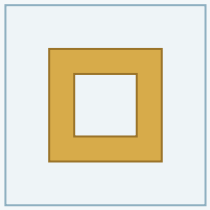
SQUARE

L = 120 nm



CIRCULAR RING

Do = 180, Di = 100



SQUARE RING

Lo = 180, Li = 100

Original dimensions are shown for reference, not as restrictions.

Allowed design parameters

Geometry	Parameters	Allowed values
Disk	Diameter D	$0 \le D \le 320$ nm
Square	Side L	$0 \le L \le 320$ nm
Circular ring	Outer Do, inner Di	$0 \le Di \le Do \le 320$ nm
Square ring	Outer Lo, inner Li	$0 \le Li \le Lo \le 320$ nm

Diameter/side sampling grid

Use n uniformly spaced values from 0 to 320 nm, including both endpoints. The geometry step is

$$\Delta = 320 / (n - 1) \quad \text{nm}$$

Each solid family has n settings. Each ring family has n(n + 1)/2 valid ordered outer/inner pairs, including zero-size and equal-size limits.

$$N_{geometry} = 2n + 2[n(n + 1)/2] = n(n + 3)$$

For every ring, sample all pairs with inner ≤ outer. No inner-diameter, outer-diameter, wall-width, or edge-gap threshold is imposed. At the boundary, some configurations coincide physically and can share one saved result.